An Attention-Based Mechanism to Combine Images and Metadata in Deep Learning Models Applied to Skin Cancer Classification

Andre G. C. Pacheco and Renato A. Krohling

***Abstract*—Computer-aided skin cancer classification systems built with deep neural networks usually yield predictions based only on images of skin lesions. Despite presenting promising results, it is possible to achieve higher performance by taking into account patient demographics, which are important clues that human experts consider during skin lesion screening. In this article, we deal with the problem of combining images and metadata features using deep learning models applied to skin cancer classification. We propose the Metadata Processing Block (MetaBlock), a novel algorithm that uses metadata to support data classification by enhancing the most relevant features extracted from the images throughout the classification pipeline. We compared the proposed method with two other combination approaches: the MetaNet and one based on features concatenation. Results obtained for two different skin lesion datasets show that our method improves classification for all tested models and performs better than the other combination approaches in 6 out of 10 scenarios.**

***Index Terms*—Convolutional neural networks, data aggregation, deep learning, skin cancer classification.**

I. Introduction

According to the World Health Organization (WHO), skin cancer represents one-third of all types of cancers diagnosed in the world [1]. Over the past decades, the incidence of this disorder has been increasing globally at a constant rate [1], [2]. In order to diagnose skin cancer, dermatologists screen the skin lesion, assess the patient clinical information, and use their experience to classify the lesion [3]. However, accurate diagnosis is challenging and depends on proper training and experience in dermoscopy [4]–[6], a non-invasive diagnostic technique that uses optic magnification to permit the visualization of morphologic features that are not visible to the naked eye [7]. Therefore, the high incidence and the lack of experts have increased the demand for computer-aided diagnosis (CAD) systems for skin cancer.

Deep neural networks (DNNs) have recently become viable for medical image analysis, including skin cancer classification [8]. In particular, Convolutional Neural Networks (CNN) are currently the standard approach to deal with this task [8]–[10]. Notable results using CNNs are reported by Steva *et al.* [11], Codella *et al.* [12], Yu *et al.* [13], among many others [3], [14]. Studies comparing the performance of deep learning models and dermatologists are reported by Haenssle *et al.* [15] and Brinker *et al.* [16]. In both works, dermatologists present similar or lower performance than the automated models. In summary, the majority of these works are based on dermoscopy images and do not take into account patient demographics.

Patient demographics – also known as patient data – are important clues that dermatologists consider during the skin lesion screening. Features like patients' age, anatomical region, cancer history, skin phototype, among many others, are common characteristics that experts pay attention to provide a more accurate clinical diagnosis. Kharazmin *et al.* [17], Liu *et al.* [18], and Pacheco and Krohling [3] proposed models to combine both skin lesion images and patient demographics. Although they report promising improvements, all of these works combine both data using feature concatenation, which may not take into account the potential relationship between metadata and the visual features extracted from the images. Recently, Li *et al.* [19] proposed the MetaNet: a multiplication-based data fusion approach that uses a sequence of 1D convolution on metadata to extract coefficients to assist visual features extracted from images in the classification task. The authors applied this method for skin cancer classification and achieved superior performance than the concatenation methodology. However, the method fails to improve melanoma classification, which is the deadliest case of skin cancer. Essentially, there is still room for improvement and better performance may be achieved using a better aggregation methodology.

In this work, we deal with the problem of combining images and metadata using deep learning models. We propose an approach named Metadata Processing Block (MetaBlock), which

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is a structure that uses metadata to support data classification by enhancing the most relevant features extracted from the images. The MetaBlock is designed to deal with minorCNNs and is agnostic to the model's architecture. In order to assess the proposed approach, we apply it to classify two different skin lesion datasets – ISIC 2019 [20] and PAD-UFES-20 [21] – and compare it with the concatenation baseline approach [3] and the MetaNet [19]. The obtained results show that our method is able to improve classification performance for both datasets and generally performs better than the other two combination approaches.

The remaining of this paper is organized as follows: In Section II, we provide an overview of image and metadata combination task; Next, in Section III, we describe the proposed approach the MetaBlock; In Section IV, we present the experimental results and a discussion about them; Lastly, we draw our conclusions.

II. Combining Image and Metadata Features

In Information Fusion (IF) the term multimodal fusion or heterogeneous fusion stands for the task of combining different types of data obtained from different sources [22]. Over the past few years, multimodal fusion has been actively applied to multimedia analysis. For example, Ortega *et al.* [23] proposed a model to combine audio, video, and text data to perform human emotion recognition. The rationale behind using such a method is that combining heterogeneous data may provide complementary information as well as increase effectiveness in a decision-making pipeline [22]. There are different strategies to perform multimodal fusion, nonetheless, the most widely used is the feature level fusion, i.e., when features extracted from each type of data are combined using a predefined methodology. This is the off-the-shelf method applied to multimedia analysis that involves image processing.

Many image processing tasks such as disease classification, face recognition, image retrieval, and object identification are challenging because the difference in the images' visual pattern is quite small. This may be caused by different factors, for example, noise, viewpoint, and data variance [24]. In order to improve performance and increase the robustness of methods that deal with these tasks, it is a common practice combine different features extracted from different algorithms or obtained from different sources of data [3], [17], [18], [25].

Several methods have been proposed to combine image features extracted from Convolutional Neural Networks (CNNs) and handcrafted features, i.e., those features extracted using traditional image processing algorithms – to extract shapes, color, or texture – or other computer vision methods such as Fisher Vector [26], Scale-Invariant Feature Transform (SIFT) [27], Speed up Robust Feature (SURF) [28], and Vector of Locally Aggregated Descriptors (VLAD) [29]. To name a few, Arroyo and Zapirian [30] and Majtner *et al.* [31] combined convolutional and hancrafted features to improve skin cancer classification, Nanni *et al.* [32] for music genre classification, and Nguyen *et al.* [33] for face detection. Other works, such as Lin *et al.* [24], Arroyo *et al.* [34], Liu *et al.* [35], and Ardila *et al.* [25], proposed the combination of image features extracted from different CNN architectures. In both cases, the most common way to combine the extracted features are through feature concatenation [32], [36], [37]. Basically, the main idea of this method is to attach all extracted features within the same structure and use it as input to another model, for example, a classifier. This is an effective and simple way to achieve feature fusion, which justifies why this method is widely used.

A slightly different problem is to combine features extracted from images with other features obtained from different sources of data. For example, in medical disease detection, we may have images and patient clinical data available to support the decision over the image [3], [18], [38]. In other words, the image is the main source of information and the extra data – usually named as metadata – provides additional information about the problem. Likewise in the previous problem, a simple concatenation is also the most common approach to combine both images and metadata [18], [39]. For example, Kharazmi *et al.* [17] proposed a feature fusion system based on concatenation and Sparse Autoencoder (SAE) to detect a specific type of skin cancer named Basal Cell Carcinoma; and Sierra *et al.* [40] also used concatenation to combine image features, extracted using two CNNs architectures, with textual metadata to perform user gender prediction.

In general, as images are high dimensional data, the number of features extracted from them is much higher than the ones extracted from metadata. In this context, a simple concatenation may not be effective. Different works have proposed approaches that, first, select/reduce the image features – using Principal Component Analysis [41] or a Neural Network, for example - and then perform the concatenation step [42]. This kind of approach assumes that by selecting the image features, the pattern within both sets of features may be easier identified by a classifier, for example. Nonetheless, as stated by Liu *et al.* [19], concatenation approaches are not able to enforce a model to focus on specific parts of the features extracted from images. Particularly, such an approach may not consider the potential of the metadata for visual tasks. Thereby, it is possible to improve performance by providing better aggregation approaches than concatenation.

III. Metadata Processing Block (MetaBlock): An Attention-Based Mechanism to Combine Multi-Source Features

In this section, we present the Metadata Processing Block (MetaBlock) an attention-based mechanism approach [43], [44] that uses the metadata to enhance the feature maps extracted from the image. The building blocks of the proposed approach are inspired by the Long Short-Term Memory (LSTM) [45] gates, which are basic composed of a single layer neural network, an activation function, and an element-wise operation.

Let us consider a classification problem in which each sample is composed of an image ($\mathbf{x}_{\text{img}}$), a group of metadata representing context information ($\mathbf{x}_{\text{meta}}$), and a label $y \in \{1, \dots, N_{\text{lab}}\}$, where N_{lab} is the number of labels. We represent this problem by the tuple $\{X_{\text{img}}, X_{\text{meta}}, Y\}$. For each type of data X_{img} and X_{meta},

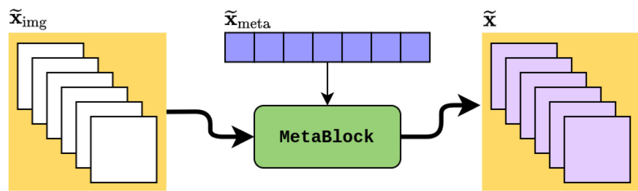


Fig. 1. A schematic diagram illustrating the main idea of the proposed combination approach. The image features are selected according to the metadata. In the end, the metadata guide the image features.

we need to define a feature extractor ψ. In this paper, we define a Convolutional Neural Network (CNN) as $\psi_{\text{img}} = g_{\text{cnn}}(\mathbf{x}_{\text{img}})$ in which we extract the last feature maps as the image features. Regarding the metadata, the extractor ψ_{meta} depends on the type of data, which is a characteristic of the classification problem. The only requirement is that ψ_{meta} return features in $\mathbb{R}^{d_{\text{meta}}}$, where d_{meta} is the dimensionality of the metadata. Therefore, we assume that $\psi_{\text{meta}}(\mathbf{x}_{\text{meta}}) \in \mathbb{R}^{d_{\text{meta}}}$. Lastly, let us define both set of features as:

$$\begin{aligned}\widetilde{\mathbf{x}}_{\text{img}} &= \psi_{\text{img}}(\mathbf{x}_{\text{img}}) \\ \widetilde{\mathbf{x}}_{\text{meta}} &= \psi_{\text{meta}}(\mathbf{x}_{\text{meta}})\end{aligned} \tag{1}$$

where $\widetilde{\mathbf{x}}_{\text{img}} \in \mathbb{R}^{k_{\text{img}} \times m_{\text{img}} \times n_{\text{img}}}$ (k_{img} and $m_{\text{img}} \times n_{\text{img}}$ are the number and order of the feature maps, respectively) and $\widetilde{\mathbf{x}}_{\text{meta}} \in \mathbb{R}^{d_{\text{meta}}}$.

At the end, our goal is to propose a method that estimates the probability of y assuming a class $c \in \{1, \ldots, N_{\text{lab}}\}$ given the image and metadata:

$$\hat{y} = p(y = c \mid \widetilde{\mathbf{x}}_{\text{img}}, \widetilde{\mathbf{x}}_{\text{meta}}) \tag{2}$$

The MetaBlock aims to guide the image according to the context information. In other words, its goal is to guide the feature maps $\widetilde{\mathbf{x}}_{\text{img}}$ according to the metadata features $\widetilde{\mathbf{x}}_{\text{meta}}$. In Fig. 1 is depicted the MetaBlock's concept. Observe that the output feature $\widetilde{\mathbf{x}}$ will have the same shape of $\widetilde{\mathbf{x}}_{\text{img}}$. However, it is modified to select the most relevant features according to the metadata features $\widetilde{\mathbf{x}}_{\text{meta}}$.

Essentially, we want the MetaBlock to work similarly to an attention mechanism. It must help the model to concentrate on more important features by incorporating the metadata knowledge into the image feature maps. To achieve this goal, we define the MetaBlock equation similar to the one used by batch normalization algorithms [46], i.e., the goal is to learn a function to scale and shift the image features according to the metadata and apply LSTM-like gates to select the most relevant features. The MetaBlock equation is defined as follows:

$$\widetilde{\mathbf{x}} = \sigma\left[\tanh\left[f_b(\widetilde{\mathbf{x}}_{\text{meta}}) \odot \widetilde{\mathbf{x}}_{\text{img}}\right] + g_b(\widetilde{\mathbf{x}}_{\text{meta}})\right] \tag{3}$$

where $\odot$ is the element-wise product, $\sigma(\cdot)$ and $\tanh(\cdot)$ are the sigmoid and hyperbolic tangent, respectively, and are designed to work as the LSTM gates. The block is based on two functions $f_b(\widetilde{\mathbf{x}}_{\text{meta}})$ and $g_b(\widetilde{\mathbf{x}}_{\text{meta}})$. Basically, these functions compose an affine transformation on the image features – similar to the batch normalization [46] strategy – which is an effective way to modify features while preserving their dimension. They can be designed as any learnable function. Nonetheless, likewise a LSTM gate, we model both functions as a single-layer neural network since it is a straightforward way to attach a non-linear function into a deep learning pipeline. Thereby, the $f_b(\widetilde{\mathbf{x}}_{\text{meta}})$ and $g_b(\widetilde{\mathbf{x}}_{\text{meta}})$ are defined as:

$$f_b(\widetilde{\mathbf{x}}_{\text{meta}}) = W_f^T \widetilde{\mathbf{x}}_{\text{meta}} + \mathbf{w}_{0_f} \tag{4}$$

$$g_b(\widetilde{\mathbf{x}}_{\text{meta}}) = W_g^T \widetilde{\mathbf{x}}_{\text{meta}} + \mathbf{w}_{0_g} \tag{5}$$

where both matrices of weights $\{W_f, W_g\} \in \mathbb{R}^{d_{\text{meta}} \times k_{\text{img}}}$ and the bias $\{\mathbf{w}_{0_f}, \mathbf{w}_{0_g}\} \in \mathbb{R}^{k_{\text{img}}}$ – recall that k_{img} is the number of feature maps extracted from the image. Therefore, each function return k_{img} coefficients, which we name modifier. Modifier coefficients may be interpreted as weights that modify feature maps in order to enhance them and to help the model concentrating on more important features.

After modifying the feature maps using the modifier coefficients, we select the most relevant features using the hyperbolic tangent and sigmoid functions gates. In other words, these gates decide which features should be let through and which one should not. From 3, we describe both gates as follows:

- **hyperbolic tangent gate**: it is the first gate of the MetaBlock and is defined by this part of the 4:

$$T_{\text{gate}} = \tanh\left[f_b(\widetilde{\mathbf{x}}_{\text{meta}}) \odot \widetilde{\mathbf{x}}_{\text{img}}\right] \tag{6}$$

  This gate controls the value of the scaling operator by adding a non-linearity that outputs values ranging from -1 to 1. Essentially, the rationale behind this gate is to increase or reduce the relevance of each feature by modifying its value to the range [-1,1], where 1 is the most relevant and -1 the opposite. The modifier coefficient responsible for the scaling learns to identify this pattern.
- **sigmoid gate**: it is the second gate of the MetaBlock and operates using the output of the previous gate T_{gate} as follows:

$$S_{\text{gate}} = \sigma\left[T_{\text{gate}} + g_b(\widetilde{\mathbf{x}}_{\text{meta}})\right] \tag{7}$$

  This gate shifts the values of the previous gate and outputs values ranging from 0 to 1. In other words, it works similarly to the previous one, however, this gate has the power to turn-off a given feature by setting it to zero. Therefore, the key idea is to output the most relevant features.

In Fig. 2 the MetaBlock structure is illustrated. In brief, we observe that the metadata is sent to the g_b and f_b functions, which will produce the modifier coefficients. Next, the feature maps are scaled by f_b, and the output is modified by the hyperbolic tangent gate. Finally, the result of the previous operation is shifted by the output of the g_b and selected by the sigmoid gate, which produces the block's output.

In Fig. 3 is illustrated a CNN schematic in which the MetaBlock is used to combine the metadata and the feature maps. Observe that the output of the MetaBlock layer is used to feed the classification layer. However, it can be used to feed any type of layer before sending them to the classifier. Also, it is important to note that this method is agnostic to the CNN architecture. As we show in the experiments, it can be easily adapted to different architectures.

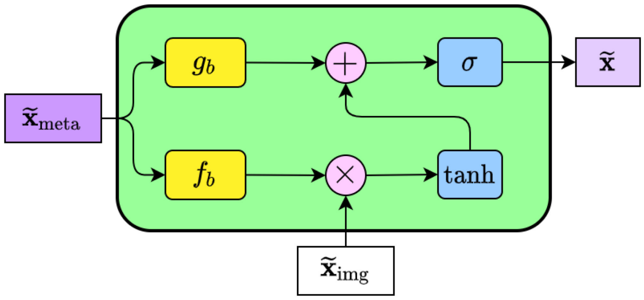


Fig. 2. The internal structure of the MetaBlock. In summary, the block learns how to enhance the image features based on the metadata features. The output features array has the same shape as the image features.

As previously described, the functions f_b and g_b are single-layer neural networks. Since the CNN is trained by using an end-to-end backpropagation algorithm, we include the weights of each function within the CNN's training parameters and perform the standard training phase. Also, we apply batch normalization to regularize the both functions.

IV. Experiments and Results

In this section, we carry out experiments to evaluate the MetaBlock's performance. We use five different CNN architectures trained on two different skin lesion datasets: ISIC 2019 [20] and PAD-UFES-20 [21]. We start this section by presenting the experiment's setup, including the dataset and the CNNs. Next, we present the results for each of the datasets. Lastly, we provide a discussion about the results.

A. Experiment's Setup

minorWe evaluate the MetaBlock using as input the last feature map layer of the following CNN architectures: EfficientNet-B4 [47], DenseNet-121 [48], MobileNet-v2 [49], ResNet-50 [50], and VGG-13 [51]. minorAll models are implemented using Pytorch.[1] and trained on two skin lesion datasets:

- **ISIC 2019** [20]: a skin lesion archive containing 25 331 public and 8238 private dermoscopy images, three patient clinical features – age, gender, and anatomical region – and eight skin lesions: Melanoma (MEL), Melanocytic Nevus (NEV), Basal Cell Carcinoma (BCC), Actinic Keratosis (ACK), Benign Keratosis (BKL), Dermatofibroma (DF), Vascular Lesion (VASC), and Squamous Cell Carcinoma (SCC).
- **PAD-UFES-20** [21]: a skin lesion dataset composed of 2298 clinical images collected from smartphone devices, 21 patient clinical features – such as age, gender, anatomical region, cancer history, skin phototype, family background, among others – and six skin lesions: Basal Cell Carcinoma (BCC), Squamous Cell Carcinoma (SCC), Actinic Keratosis (ACK), Seborrheic Keratosis (SEK), Melanoma (MEL), and Nevus (NEV).

The key differences between the datasets are the type of image – dermoscopy and clinical – and the number of patient clinical features. Essentially, dermoscopy images provide more details about the skin surface and they are not/less affected by illumination or camera resolution compared to the clinical ones.

We compare the MetaBlock approach with the models without using metadata, the baseline concatenation method [3], and the MetaNet [19]. All combination methods are attached to the original architecture of the five CNN models employed in this experiment. All models were pre-trained on ImageNet [52] minor– with the weights gathered from torchvision API – and fine-tuned on ISIC 2019 and PAD-UFES-20 for 150 epochs using SGD optimizer with a learning rate equal to 0.001, momentum equal to 0.9, and weight decay equal to 0.001. The learning rate is reduced by a rate of 0.1 if the model does not improve for 10 consecutive epochs. In addition, we used early stopping if the model does not improve for 15 consecutive epochs. As the dataset is imbalanced, i.e., the samples are not represented equally among the labels, we applied the weighted cross-entropy as the loss function in which the weights are determined according to the labels' frequency. All images were resized to 224×224 and we applied data augmentation using common image processing operations: horizontal and vertical flips, adjustments in brightness, contrast, and, saturation, image scaling, and random noise [14], [53].

For all experiments, we applied 5-fold cross-validation stratified by the label's frequency for assessing the effectiveness of the models. To measure the performance, we computed the average and standard deviation of the following metrics: accuracy (ACC), balanced accuracy (BACC), the aggregated area under the curve (AUC). As the dataset is imbalanced, we consider BACC as the main metric. In order to compare the experiments results, we perform the non-parametric Friedman test following by the Wilcoxon test (if applicable), using $p_{\text{value}} = 0.05$ [54]. We also perform the A-TOPSIS [55], a straightforward method that ranks a group of evaluated algorithms according to their performance. In brief, the method takes as input a matrix where each line represents an algorithm within the group and is composed of the mean and standard deviation of a given metric. Through a similarity metric, the method determines a score for each algorithm which is used to rank them.

B. Metadata Pre-Processing

For both datasets we use the patient clinical features (demographics) as the metadata to support the image classification. As previously stated, ISIC 2019 dataset contains only three clinical features, while the PAD-UFES-20 contains 21. In both cases, we have features represented by a number, for example, age and diameter; by strings, such as anatomical region and gender; and by a boolean, for instance, cancer history and if the skin lesion hurts. We describe how we handle each type of feature as follows:

- *Numerical features:* when the feature is represented by a number (integer or float), we do not apply any transformation and use them as they are. In fact, we tried to normalize these features between [0,1]; however, it does not present any change in the final result.
- *Boolean features:* for this type of features, we just represent `True` as 1 and `False` as 0.

[1]Code is available on https://github.com/paaatcha/MetaBlock

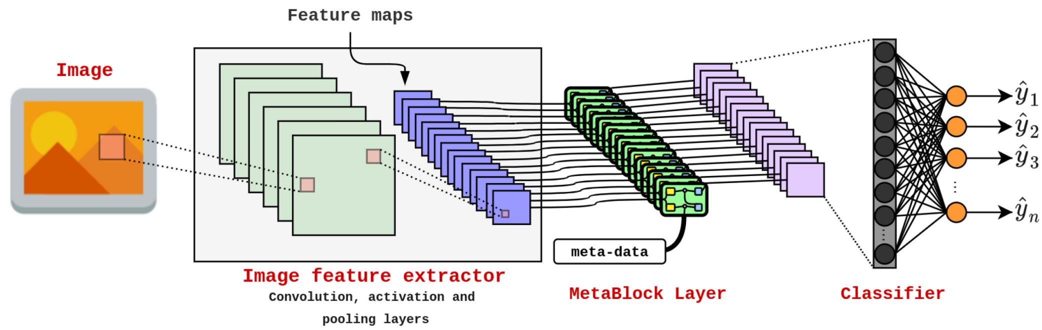


Fig. 3. An illustration of a MetaBlock layer attached to a CNN model. Observe that the metadata are used to identify the most relevant features from the feature maps before sending them to the classifier.

- *String/Categorical features*: When the feature is represented by a string, it has a known number of options. For example, the gender attribute may assume two values: `male` or `female`. In order to use these features as metadata, we need to apply a feature extractor ψ_{meta} to transform the categorical data into scalar numbers. Thereby, we applied the one-hot encoding strategy [56] to encode the categorical attributes present in the patient demographics into an array $\widetilde{\mathbf{x}}_{\text{meta}} \in \mathbb{R}^{d_{\text{meta}}}$ – particularly, this is the same strategy employed by Li *et al.* [19] to use MetaNet. For example, for the gender attribute, the employed strategy may assume [1,0], [0,1], or [0,0] if gender is `male`, `female`, or if this information is missing, respectively. We applied the one-hot encoding strategy mainly because the data do not demand a fancy strategy to encode it properly, i.e., the values that each attribute assumes are well defined and are within a set of known values; the method is simple to implement; and it provides a way to map the missing data.

For the ISIC 2019 dataset, the one-hot encoding strategy transforms the three categorical features into an array $\mathbf{x}_{\text{meta}} \in \mathbb{R}^{11}$, where $d_{\text{meta}} = 11$ is the number of features. On the other hand, for the PAD-UFES-20 dataset, the same strategy encodes the 21 categorical features into an array $\mathbf{x}_{\text{meta}} \in \mathbb{R}^{81}$, where $d_{\text{meta}} = 81$ is also the number of features. In both case, the numerical features are used as metadata to support the image skin lesion classification.

C. Experiment Results - ISIC 2019 Dataset

We now present the results obtained for the ISIC 2019 dataset with and without considering the metadata. In Table I is presented the results, in terms of mean and standard deviation, for each CNN model without using metadata, for the models using the concatenation approach, for MetaNet, and for the models using the MetaBlock. To ease the visualization, in Table II is presented the performance only in terms of BACC for each method.

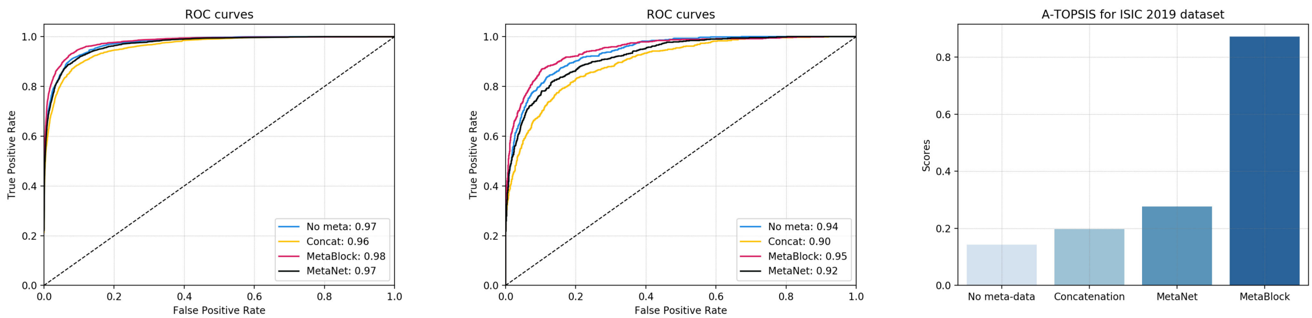
As we can note from Tables I and II, the use of the patient clinical features to support the image classification provides a fair performance improvement in terms of BACC, in particular when the MetaBlock approach is applied, which improved performance for 3 out of 5 CNN models. As the dataset is imbalanced, the ACC is slightly higher than the BACC for all models, which is expected. To compare the four methods, we performed the Friedman and Wilcoxon tests, considering the BACC metric as the tests input. The Friedman test returned $p_{value} = 0.0011$. Thus we performed the pairwise Wilcoxon test, which is described in Table III. As we can see, the test indicates that the MetaBlock approach is statistically different from the remaining ones. We also performed the A-TOPSIS to

TABLE I
PERFORMANCE OF THE CNN MODELS FOR THE ISIC 2019 DATASET WITH AND WITHOUT CONSIDERING THE METADATA

Model	ACC	BACC	AUC
No metadata			
EfficientNet-B4	0.776 ± 0.023	0.755 ± 0.013	0.956 ± 0.004
DenseNet-121	0.774 ± 0.029	0.755 ± 0.024	0.961 ± 0.006
MobileNet-V2	0.763 ± 0.008	0.742 ± 0.009	0.961 ± 0.001
ResNet-50	0.767 ± 0.036	0.744 ± 0.035	0.959 ± 0.007
VGGNet-13	0.722 ± 0.018	0.701 ± 0.016	0.948 ± 0.004
Concatenation			
EfficientNet-B4	0.784 ± 0.004	0.768 ± 0.015	0.96 ± 0.002
DenseNet-121	0.738 ± 0.012	0.737 ± 0.01	0.952 ± 0.002
MobileNet-V2	0.716 ± 0.009	0.723 ± 0.016	0.946 ± 0.004
ResNet-50	0.729 ± 0.013	0.726 ± 0.013	0.948 ± 0.005
VGGNet-13	0.724 ± 0.015	0.729 ± 0.013	0.949 ± 0.004
MetaBlock			
EfficientNet-B4	0.807 ± 0.008	0.762 ± 0.011	0.962 ± 0.003
DenseNet-121	0.800 ± 0.011	0.769 ± 0.013	0.965 ± 0.002
MobileNet-V2	0.777 ± 0.006	0.76 ± 0.004	0.958 ± 0.001
ResNet-50	0.804 ± 0.006	0.771 ± 0.011	0.966 ± 0.002
VGGNet-13	0.753 ± 0.031	0.74 ± 0.015	0.955 ± 0.007
MetaNet			
EfficientNet-B4	0.766 ± 0.019	0.756 ± 0.012	0.959 ± 0.004
DenseNet-121	0.725 ± 0.020	0.723 ± 0.016	0.949 ± 0.005
MobileNet-V2	0.742 ± 0.019	0.731 ± 0.015	0.955 ± 0.006
ResNet-50	0.753 ± 0.030	0.746 ± 0.022	0.956 ± 0.008
VGGNet-13	0.767 ± 0.020	0.746 ± 0.013	0.959 ± 0.005

TABLE II
COMPARING THE PERFORMANCE OF THE ALL METHODOLOGIES IN TERMS OF BACC FOR THE ISIC 2019 DATASET. IN BOLD IS HIGHLIGHTED THE HIGHEST AVERAGE FOR EACH MODEL

Model	No metadata	Concatenation	MetaBlock	MetaNet
EfficientNet-B4	0.755 ± 0.013	$\mathbf{0.768 \pm 0.015}$	0.762 ± 0.011	0.756 ± 0.012
DenseNet-121	0.755 ± 0.024	0.737 ± 0.01	$\mathbf{0.769 \pm 0.013}$	0.723 ± 0.016
MobileNet-v2	0.742 ± 0.009	0.723 ± 0.016	$\mathbf{0.76 \pm 0.004}$	0.731 ± 0.015
ResNet-50	0.744 ± 0.035	0.726 ± 0.013	$\mathbf{0.771 \pm 0.011}$	0.746 ± 0.022
VGGNet-13	0.701 ± 0.016	0.729 ± 0.013	0.74 ± 0.015	$\mathbf{0.745 \pm 0.013}$



(a) Macro average (b) Melanoma (c) A-TOPSIS

Fig. 4. The Macro average (a) and melanoma ROC (b) curves for no metadata, concatenation, MetaBlock, and MetaNet approaches considering the ResNet-50 performance on ISIC 2019 dataset. In (c) is depicted the A-TOPSIS rank for each approach.

TABLE III
THE RESULT OF THE WILCOXON PAIRWISE TEST FOR ALL METHODS PERFORMED FOR ISIC 2019 DATASET

Pair	p_{value}
No metadata - Concatenation	0.502
No metadata - MetaBlock	0.002
No metadata - MetaNet	0.773
MetaBlock - Concatenation	0.001
MetaBlock - MetaNet	0.017
MetaNet - Concatenation	0.343

TABLE IV
PERFORMANCE OF EACH COMBINATION APPROACH USING THE RESNET-50 MODEL FOR THE ISIC 2019 PRIVATE TEST SET

Method	ACC	BACC	AUC Macro
No metadata	0.913	0.522	0.865
Concatenation	0.909	0.517	0.860
MetaBlock	0.910	0.550	0.866
MetaNet	0.909	0.523	0.866

rank the three methods. The rank is presented in Fig. 4(c) and it is in accordance with the statistical test, i.e., the MetaBlock approach has a higher rank score and the three remaining methods achieve quite similar values.

To get a gist of the AUC metric, in Fig. 4 is shown the ROC curves, considering the ResNet-50 model,[2] with no metadata, concatenation, MetaNet, and the MetaBlock approaches considering macro average and melanoma, the deadliest type of skin cancer. As we can see, both ROC curves for the MetaBlock approach are above the remaining methods, which is in line with the previous results.

To conclude this experiment, we evaluate the combination approaches using the private test partition of the ISIC 2019. As the ground truth for this test set is not publicly available, we evaluate the algorithms using the ISIC live challenge platform.[3] This platform limits the number of submissions per week; thus, we also performed each method using the ResNet-50 model. In Table IV is presented three performance metrics that the platform makes available for us – the BACC is the one used to rank the submissions. As we can see, the MetaBlock approach achieves the highest BACC, even though the remaining metrics are quite close. The MetaNet is slightly better than the model without using metadata, and the concatenation is slightly worse. It is worth noting that this performance is achieved for a single model without using any external data to train the model. Also, this partition has a UNK category that we are not handling in these submissions because it is beyond the scope of this paper. This helps to explain the difference of the performance obtained for public and private partitions.

D. Experiment Results - PAD-UFES-20 Dataset

In this part of the experiment, we compare the deep learning models' performance with and without considering the metadata for the PAD-UFES-20 dataset [21]. In Table V is presented the results, in terms of mean and standard deviation, for each method. To ease the visualization, in Table VI is summarized the performance only in terms of BACC for each method. As we can note from the tables, there is also a notable improvement for all metrics when the metadata is included in the classification process. Observing the models' performance when the metadata is employed, we note that they present performances varying from 71.7% to 77% in terms of BACC. Quantitatively, there is an improvement of at least 8.2% in BACC when MetaBlock is applied. Similar to ISIC 2019 dataset, the MetaBlock presents a more stable performance compared to the concatenation and MetaNet. However, the improvement was much higher, in particular, because we have access to a more valuable 21 patient clinical features instead of only 3. In order to investigate this difference, we repeated the same experiment using PAD-UFES-20, but using only the same three metadata available in ISCI 2019: age, gender, and anatomical region. In Table VII is shown the performance for each combination approach in terms of BACC. Comparing to the results presented in Table VI, the improvement is much lower – it is the same magnitude of the improvement obtained for ISIC 2019 –, which indicates that the additional metadata available in this dataset is crucial to achieve the previous performance.

[2] As we would have 10 ROC curves, we decided to present these plots only for the ResNet-50 because it is a fast and a quite common model in Deep Learning, including skin cancer classification; and it presents a fair performance for our experiments

[3] https://challenge2019.isic-archive.com/

TABLE V
PERFORMANCE OF THE CNN MODELS FOR THE PAD-UFES-20 DATASET WITH AND WITHOUT CONSIDERING THE METADATA

No metadata			
Model	**ACC**	**BACC**	**AUC**
EfficientNet-B4	0.656 ± 0.027	0.64 ± 0.029	0.911 ± 0.006
DenseNet-121	0.636 ± 0.055	0.64 ± 0.042	0.893 ± 0.02
MobileNet-V2	0.655 ± 0.016	0.637 ± 0.018	0.898 ± 0.01
ResNet-50	0.616 ± 0.051	0.651 ± 0.05	0.901 ± 0.007
VGGNet-13	0.709 ± 0.019	0.654 ± 0.022	0.901 ± 0.003
Concatenation			
Model	**ACC**	**BACC**	**AUC**
EfficientNet-B4	0.765 ± 0.032	0.758 ± 0.012	0.945 ± 0.005
DenseNet-121	0.742 ± 0.035	0.747 ± 0.019	0.932 ± 0.005
MobileNet-V2	0.738 ± 0.026	0.741 ± 0.017	0.927 ± 0.007
ResNet-50	0.741 ± 0.014	0.728 ± 0.029	0.929 ± 0.006
VGGNet-13	0.712 ± 0.035	0.72 ± 0.018	0.929 ± 0.002
MetaBlock			
Model	**ACC**	**BACC**	**AUC**
EfficientNet-B4	0.748 ± 0.018	0.77 ± 0.016	0.944 ± 0.004
DenseNet-121	0.723 ± 0.037	0.746 ± 0.039	0.931 ± 0.006
MobileNet-V2	0.724 ± 0.016	0.754 ± 0.014	0.938 ± 0.005
ResNet-50	0.735 ± 0.013	0.765 ± 0.017	0.935 ± 0.004
VGGNet-13	0.728 ± 0.022	0.736 ± 0.018	0.933 ± 0.002
MetaNet			
Model	**ACC**	**BACC**	**AUC**
EfficientNet-B4	0.744 ± 0.014	0.737 ± 0.017	0.931 ± 0.007
DenseNet-121	0.745 ± 0.022	0.745 ± 0.029	0.932 ± 0.008
MobileNet-V2	0.700 ± 0.013	0.717 ± 0.020	0.922 ± 0.005
ResNet-50	0.732 ± 0.054	0.742 ± 0.019	0.936 ± 0.006
VGGNet-13	0.749 ± 0.037	0.754 ± 0.033	0.937 ± 0.011

TABLE VI
COMPARING THE MODELS' PERFORMANCE IN TERMS OF BACC FOR EACH METHOD FOR THE PAD-UFES-20 DATASET. IN BOLD IS HIGHLIGHTED THE HIGHEST AVERAGE BACC FOR EACH MODEL

Model	No metadata	Concatenation	MetaBlock	MetaNet
EfficientNet-B4	0.64 ± 0.029	0.758 ± 0.012	$\mathbf{0.77 \pm 0.016}$	0.737 ± 0.017
DenseNet-121	0.64 ± 0.042	$\mathbf{0.747 \pm 0.019}$	0.746 ± 0.039	0.745 ± 0.029
MobileNet-V2	0.637 ± 0.018	0.741 ± 0.017	$\mathbf{0.754 \pm 0.014}$	0.717 ± 0.020
ResNet-50	0.651 ± 0.05	0.728 ± 0.029	$\mathbf{0.765 \pm 0.017}$	0.742 ± 0.019
VGGNet-13	0.654 ± 0.022	0.72 ± 0.018	0.736 ± 0.018	$\mathbf{0.754 \pm 0.033}$

TABLE VII
COMPARING THE MODELS' PERFORMANCE IN TERMS OF BACC FOR EACH METHOD FOR THE PAD-UFES-20 DATASET CONSIDERING ONLY THE SAME THREE FEATURES AVAILABLE IN ISIC 2019. IN BOLD IS HIGHLIGHTED THE HIGHEST AVERAGE BACC FOR EACH MODEL

Model	No metadata	Concatenation	MetaBlock	MetaNet
EfficientNet-B4	0.640 ± 0.029	0.667 ± 0.038	$\mathbf{0.700 \pm 0.025}$	0.672 ± 0.070
DenseNet-121	0.640 ± 0.042	0.614 ± 0.036	$\mathbf{0.641 \pm 0.021}$	0.625 ± 0.036
MobileNet-v2	0.637 ± 0.018	0.656 ± 0.043	$\mathbf{0.682 \pm 0.029}$	0.662 ± 0.038
ResNet-50	$\mathbf{0.651 \pm 0.05}$	0.608 ± 0.019	0.639 ± 0.013	0.626 ± 0.036
VGGNet-13	0.654 ± 0.022	0.611 ± 0.033	0.633 ± 0.024	$\mathbf{0.662 \pm 0.047}$

In this experiment, we also selected the ResNet-50 for further analysis. In Fig. 5 is depicted the confusion matrix for each combination approach considering the selected model. The confusion matrices show an interesting result, in general, the metadata helped to improve the diagnostic rates for ACK, MEL, NEV, and SEK. However, it increases miss-classification between SCC and BCC. It happens because both lesions are quite similar and share almost the same values of clinical features. This behavior is also observed in the t-SNE [57] plots depicted in Fig. 6. As we can note, the combination approaches improve the clustering of the samples; however, they still struggle to differentiate the non-pigmented skin lesions. In fact, distinguishing SCC from BCC is a challenging task even for experts using a dermatoscope.

TABLE VIII
THE RESULT OF THE WILCOXON PAIRWISE TEST FOR ALL METHODS

Pair	p_{value}
No metadata - Concatenation	$\sim 10^{-5}$
No metadata - MetaBlock	$\sim 10^{-5}$
No metadata - MetaNet	$\sim 10^{-5}$
MetaBlock - Concatenation	0.008
MetaBlock - MetaNet	0.04
MetaNet - Concatenation	0.81

Nonetheless, this confusion is not quite a problem, since both are skin cancer and need to be biopsied. The real problem is confusing them with ACK, which is just a minor skin disease that is treated without a surgical process. For MEL, the deadliest case of skin cancer, the MetaBlock approach achieves maximum performance, while the concatenation and MetaNet do no present an improvement compared to the models without using metadata.

We also performed the Friedman and Wilcoxon tests, considering the BACC metric, to compare the performances of the method. The Friedman test returned $p_{\text{value}} \approx 5 \times 10^{-11}$. Thus, we performed the Wilcoxon test, in which the p values are presented in Table VIII. As we can see, the test returns $p_{\text{value}} < 0.05$ for all pairwise comparisons, except the pair MetaNet-Concatenation. In other words, it means all methods are statistically different from each other, except the MetaNet and Concatenation approaches. Based on the results presented throughout this section, we may conclude that the MetaBlock performs better than the concatenation and MetaNet approaches. This assumption is also confirmed by the A-TOPSIS rank, depicted in Fig. 7, which shows the MetaBlock with a higher rank score while the concatenation and MetaNet have quite close values.

E. Discussion

The results presented in the previous section indicate that combining metadata and patient demographics may improve the performance of CNN models for skin cancer classification. However, the results also show the improvement depend on the combination method. For the ISIC 2019 dataset, the statistical test indicates the concatenation and MetaNet approaches do not present differences from the models without using the metadata. It happens because these methods present better performances than the no metadata one only for some models – for example, MetaNet presents higher performance only for EfficientNet-B4 and VGGNet-13. On the other hand, the MetaBlock is more stable and presents an average improvement of over 1% in balanced accuracy. Considering only the BACC, the MetaBlock approach presents the best performance for 3 out of 5 models.

Considering the PAD-UFES-20 dataset, all three combination methods present higher performance than the models without using the metadata. Similar to the ISIC 2019 dataset, the MetaBlock approach achieved the best performance, in terms of BACC, for 3 out of 5 models. The statistical test showed that the proposed method provides better results than two other combination methods. In addition, the MetaBlock improved

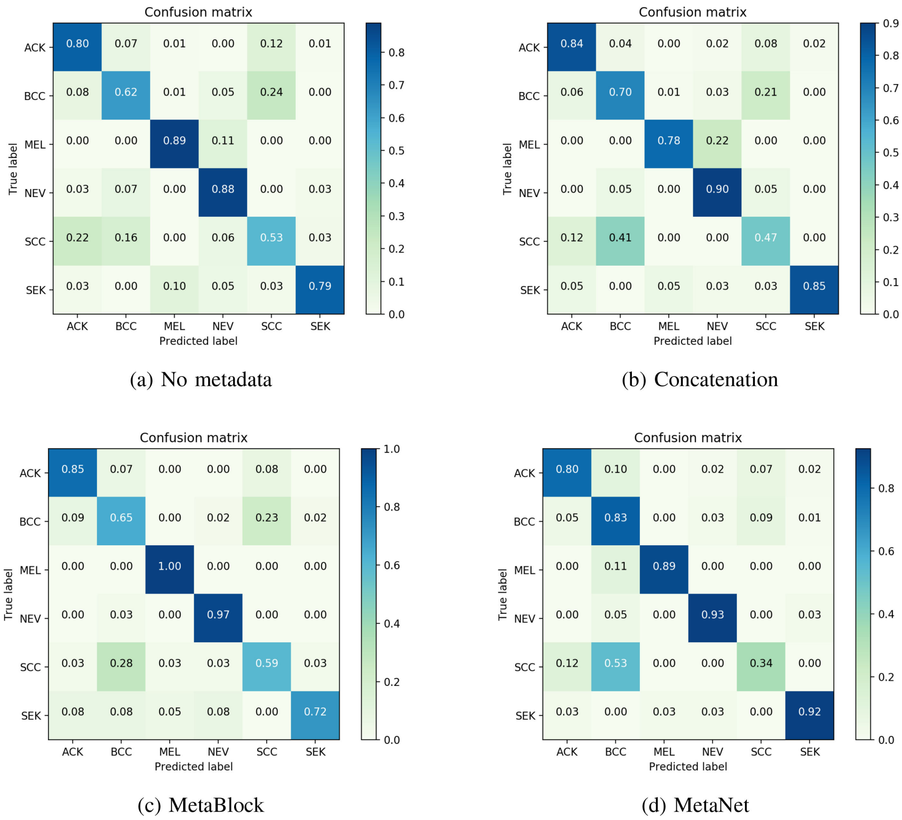


Fig. 5. The confusion matrix for each combination approach considering the ResNet-50 model.

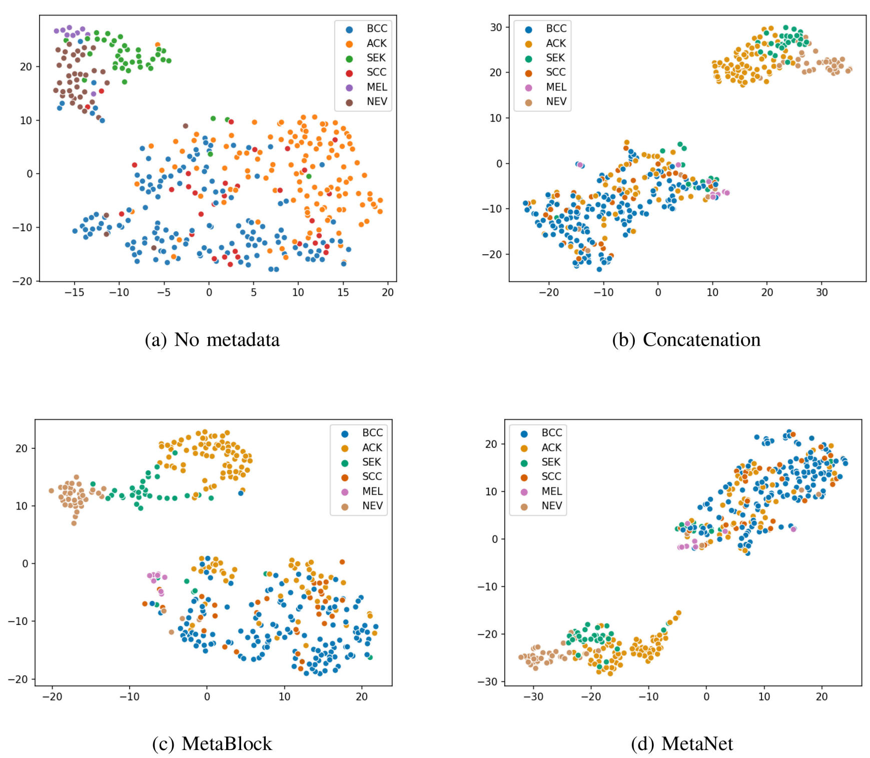


Fig. 6. The t-SNE visualization for each combination approach considering the ResNet-50 model.

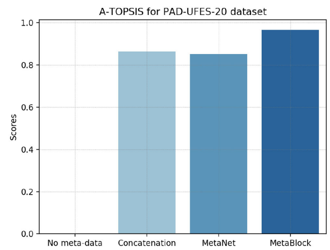


Fig. 7. The A-TOPSIS rank for each approach considering the BACC metric for the PAD-UFES-20 dataset.

melanoma classification, which is very desired for any skin cancer CAD system.

From the comparison table II and VI, the lowest performance in terms of BACC is achieved by the VGG-13 architecture. Compared to the remaining architectures, the VGG-13 is the one that outputs the higher number of feature maps ($512 \times 7 \times 7$). It suggests that our method may be less effective when the network outputs a large number of feature maps. More recent models, such as EfficientNet-B4 and ResNet-50, introduced new mechanisms to allow the addition of several lower convolutional layers to better process the image. As a result, these models output fewer feature maps in the last layer, which seems to benefit our proposed approach.

Observing the results for each dataset, we notice that the metadata provided much higher performance for the PAD-UFES-20 than to the ISIC 2019 dataset. In addition, the concatenation approach does not work well for ISIC. As we show in the experiments, this happens because the ISIC 2019 dataset has only three metadata attributes while the PAD-UFES-20 has 21. In addition, we note that the concatenation approach cannot handle it properly. Conversely, the MetaBlock approach could handle this issue well and presented a higher performance compared to all other approaches.

Li *et al.* [19] analyzed the MetaNet only in terms of recall over the labels, which also shows poor performance for melanoma classification. In addition, the authors do not provide any statistical test or another method to further compare their method with the no metadata and concatenation approaches. In this experiment, we show that MetaNet's performance depends on CNN architecture and, most of the time, it is similar to the concatenation approach.

To conclude, when we attach the MetaBlock to a CNN model it increases the model's size; however, it is almost insignificant. Considering the experiments carried out for PAD-UFES-20 – the one with with more metadata and, consequently, more data to fuse – the most impacted model is the VGG-13. We increase the number of trainable parameters by 0.85% – from 9 611 592 to 9 693 126 –, which is not significant in terms of training time. Considering the remaining models, it increases their size in 0.05%, 0.08%, 0.3%, and 0.05%, for EfficientNet-B4, DenseNet-121, MobileNet-V2, and ResNet-50, respectively.

V. Conclusion

In this section, we present the Metadata Processing Block (MetaBlock), an attention-based mechanism approach that uses the metadata to enhance the feature maps extracted from images in order to improve data classification. In order to evaluate the MetaBlock performance, we applied it to five different Convolutional Neural Network (CNN) trained on two different skin lesion dataset: ISIC 2019 and PAD-UFES-20. Also, we compared the obtained results with the models without using metadata, using a concatenation baseline, and the MetaNet. In brief, our model achieves the best performance, in terms of balanced accuracy, for 3 out of 5 CNN models for both datasets. Also, it is more stable than the other two combination methods and the statistical test indicates it generally performs better for both datasets.

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